BlOOMAR: AN AUGMENTED REALITY BOUQUET CUSTOMIZATION AND VISUALIZATION APP USING MARKERLESS AR TRACKING

A Thesis Presented to the Faculty of

Tanauan Institute, Inc.

Tanauan City, Batangas

In Partial Fulfillment of the Requirements for the

Degree of Bachelor of Science in Computer Science

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March 2026

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ACKNOWLEDGEMENT

The researchers would like to extend their profound gratitude and sincerest appreciation to the following who extended their kind assistance for the successful completion of this work.

Honorable Congresswoman Maria Theresa "Maitet" V. Collantes and Dr. Nelson "Sonny" Perez Collantes, and Collantes family for the scholarship granted to be able to finish this course;

Mr. Mark Andrew Rimas, the statistician, for his diligence and patience in formulating the statistical treatment;

To **Dr. Nelia P. Mananguit**, Dean of Studies, for endorsing the developers and allowing them to gather information from the Institution.

First and foremost, the developers would like to convey their heartfelt gratitude to their adviser **Ms. Jacinta O. Precilla** for his constant support and wise counsel. His experience in this matter has been critical in defining the course of the study and pushing the researchers to achieve excellence.

To **Mr. Bernardo M. Gonzales, Mr. Michael A. Velasquez** and **Engr. Maria Theresa B. Prenda,** whose experience in computer science gave a fresh viewpoint and improved the depth of this study. Their mentoring has been a guiding force for the researchers, and they are thankful for the time and effort they have put into my academic development.

And to the **Almighty God.,** source of grace and wisdom, love and enlightenment in making this study a reality**.**

To the **co-developers**, whose participation and devotion enriched the process of bringing this project to life. The knowledge, ideas, and shared enthusiasm for the topic all contributed greatly to the breadth and quality of the work.

To their **family** for their constant support and encouragement during the process of finishing this thesis. Their love, understanding, and patience have been the pillars that have kept them going during the difficult times.

J.M.M.A
E.C.Q

DEDICATION

We dedicate this thesis to our families and loved ones,

who have provided us with unwavering support, endless encouragement, and love.

Their belief in us has been the driving force behind our success.

To our professors and mentors,

whose guidance, wisdom, and expertise have been invaluable.

We are grateful for their patience and unwavering dedication to our academic pursuits.

To our colleagues and classmates,

who have been there for us through thick and thin, encouraging us every step of the way.

Their support and friendship have made this journey worthwhile.

*Finally, To **Almighty God**,*

our Creator, and Guide, who has blessed us with the gift of life.

Your grace and mercy have been our constant source of strength and inspiration throughout this journey.

We are grateful for your guidance, wisdom, and love, which have sustained us in moments of doubt and uncertainty.

J.M.M.A

E.C.Q

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CHAPTER I

INTRODUCTION

Flowers play a significant role in expressing emotions and enhancing special occasions such as weddings, birthdays, anniversaries, and Valentine's Day. However, clients often struggle to visualize how different flowers, colors, and arrangement styles will look once combined into a final bouquet. This difficulty may lead to misunderstandings between clients and florists, resulting in design revisions, additional costs, and dissatisfaction.

With the advancement of augmented reality (AR) technology, digital objects can now be overlaid into real-world environments through mobile devices. AR allows users to preview and interact with virtual objects in real time, making it a valuable tool for design visualization. Bloom AR: An Augmented Reality Bouquet Customization and Visualization App was developed to address the challenge of bouquet conceptualization by enabling users to design, customize, and preview floral arrangements directly within their physical surroundings.

Bloom AR incorporates marker less AR tracking supported by feature-based detection and real-time rendering algorithms. These technologies allow the system to detect flat surfaces, anchor virtual bouquets without printed markers, and maintain object stability

during user movement. The application also includes a real-flower detection feature that captures the visual characteristics of a physical flower and enables digital duplication within the system.

By integrating bouquet customization tools with AR visualization, Bloom AR transforms traditional floral planning into an interactive, user-centered, and technology-driven experience.

Project Context

BloomAR is an augmented reality bouquet customization and visualization system designed to assist users in creating floral arrangements through an interactive and realistic digital platform. The system provides four main bouquet categories for different events: wedding, birthday, anniversary, and Valentine's Day. Each category includes flower types, color options, ribbons, and arrangement styles that users can freely explore and modify.

The system allows users to customize bouquets by selecting and combining various floral elements based on their personal preference or event requirements. After designing, BloomAR generates a three-dimensional bouquet model that can be viewed in the user's real surroundings using AR technology. Through the mobile device camera, users can rotate, resize, reposition, and evaluate the bouquet from different angles, giving them a clear and accurate visualization of how the arrangement will appear in an actual setting.

A key enhancement of the system is its real-world flower detection feature. When a user points their device at a physical flower, BloomAR identifies its general appearance,

including its shape and color. The captured flower becomes a reference model that can be duplicated inside the system. This means that even if only a few flowers exist in the real world, BloomAR allows users to increase the number of flowers virtually and customize them further. They may modify the size, color variation, quantity, and positioning of the duplicated flowers. This feature provides flexibility for users who want to experiment with designs beyond what is physically available.

BloomAR aims to support customers, florists, and event coordinators by combining creativity with technology. Through real-time AR visualization and customizable bouquet design tools, the system reduces uncertainty, eliminates guesswork, and enhances decision-making during floral planning. The system serves as an accessible, user-friendly, and modern tool for creating visually appealing bouquet concepts before actual production.

Purpose and Description

The purpose of BloomAR is to provide an interactive mobile application that assists users in designing and visualizing personalized flower bouquets for different special occasions. The system is intended for clients, florists, and event organizers who need a clear and realistic preview of floral arrangements before they are physically created. Using augmented reality, Bloom AR helps reduce uncertainty during the planning process and supports better communication between clients and floral designers. Bloom AR allows users to choose from four event categories: Wedding, Birthday,

Anniversary, and Valentine's Day. Each category contains a selection of flower types, color themes, ribbons, and bouquet styles suited for the occasion. Users may freely combine these elements to create a bouquet that matches their preferences or the concept of their event.

A key component of the system is its ability to detect a real flower when viewed through the device's camera. The system interprets the appearance of the flower and allows the user to use it as part of their digital bouquet. BloomAR makes it possible to duplicate the detected flower and increase its quantity inside the application. Even if only a few flowers are present in real life, the system can create additional virtual versions that the user can modify and arrange.

This provides greater flexibility and creativity, especially during moments when physical samples are limited. Once the bouquet design is completed, users can view the arrangement in their actual environment through AR mode. The system enables them to rotate, resize, reposition, and examine the bouquet from multiple angles. The design may also be saved or captured for reference or further revision.

Overall, Bloom AR serves as a modern tool that combines creativity and technology. It provides a realistic, convenient, and user-friendly platform for conceptualizing bouquet designs, helping users make informed decisions and visualize their ideas before any physical arrangement is made.

Objective of the Study

General Objective

The general objective of this study is to develop BloomAR, a mobile application that allows users to design, modify, and visualize flower bouquets using augmented reality. The system aims to provide a realistic and interactive environment where users can create bouquet arrangements and view them within their physical surroundings.

Specific Objectives

1. Create a mobile application that offers bouquet categories suited for Weddings, Birthday, Anniversary, and Valentine's Day occasions.
2. Enable users to customize bouquet elements such as flower types, color variations, ribbon styles, and arrangement layouts.
3. Integrate AR technology that displays three-dimensional bouquets directly into real environments through the device camera.
4. Provide interactive controls that allow users to rotate, resize, and reposition the bouquet for accurate visualization.
5. Incorporate a real-flower detection feature that captures a physical flower and allows the user to duplicate and modify it inside the system for extended customization.
6. All2ow users to save or capture their finished bouquet designs for reference, consultation, or sharing.

Significance of the Study

The significance of this study lies in the fact that BloomAR enhances the planning, presentation, and visualization of bouquet designs. By using augmented reality and customizable digital elements, the system helps users overcome the common difficulty of imagining how different flowers, colors, and arrangements will look when combined. Through this approach, BloomAR contributes to a more accurate and convenient design process for various individuals and groups involved in floral arrangements.

For florists and event planners. BloomAR provides an efficient way for florists and planners to present bouquet concepts to clients. The AR features help communicate design ideas clearly, allowing professionals to instantly adjust flower quantity, arrangement style, and color combinations based on client feedback. The system reduces errors and saves time usually spent on creating multiple physical samples.

For Clients and Customers. Clients are able to design and visualize bouquets tailored to their preferences without relying solely on imagination. The AR preview gives them a realistic view of how the arrangement will appear during special events. The ability to duplicate real flowers in the app also allows customers to experiment with different bouquet styles even when only a few actual samples are available.

For Students and Researchers. The system serves as a valuable reference for those studying mobile development, AR applications, computer vision, and human-computer interaction. BloomAR demonstrates how various technologies can be integrated to solve practical design challenges.

For Developers. Developers can use the system as an example of implementing real-world object detection combined with AR visualization. The project shows how digital content can enhance physical objects and provide expanded creative functionality.

For the Researchers. The study strengthens the technical and analytical skills of the researchers by allowing them to apply their knowledge in AR, 3D modeling, mobile development, and system design. It also exposes them to the processes of identifying user needs, evaluating system performance, and solving real-world design issues.

Scope and Limitation of the Study

The study focuses on the development of Bloom AR, a mobile application designed to assist users in creating customized flower bouquets using augmented reality. The system allows users to choose from four event categories, select flower types, apply color variations, add ribbons, and arrange these elements into a personalized bouquet. It supports the visualization of three-dimensional bouquets within real environments through the device camera and provides interactive controls for rotating, resizing, and repositioning the bouquet. Bloom AR also features a real-flower detection function that enables the system to capture a physical flower and generate virtual duplicates for extended customization.

Despite these capabilities, the system includes only a limited collection of flower and ribbon designs and does not allow users to upload their own flower models. The real flower detection function captures only general visual characteristics and may not identify exact species or textures. The system does not perform advanced species-level flower

classification, as detection is based on general visual features rather than deep learning classification methods. The quality of AR visualization may be affected by lighting, camera performance, and device hardware. In addition, the system does not support online purchasing or ordering, and some features may require an internet connection to operate efficiently.

Conceptual Framework

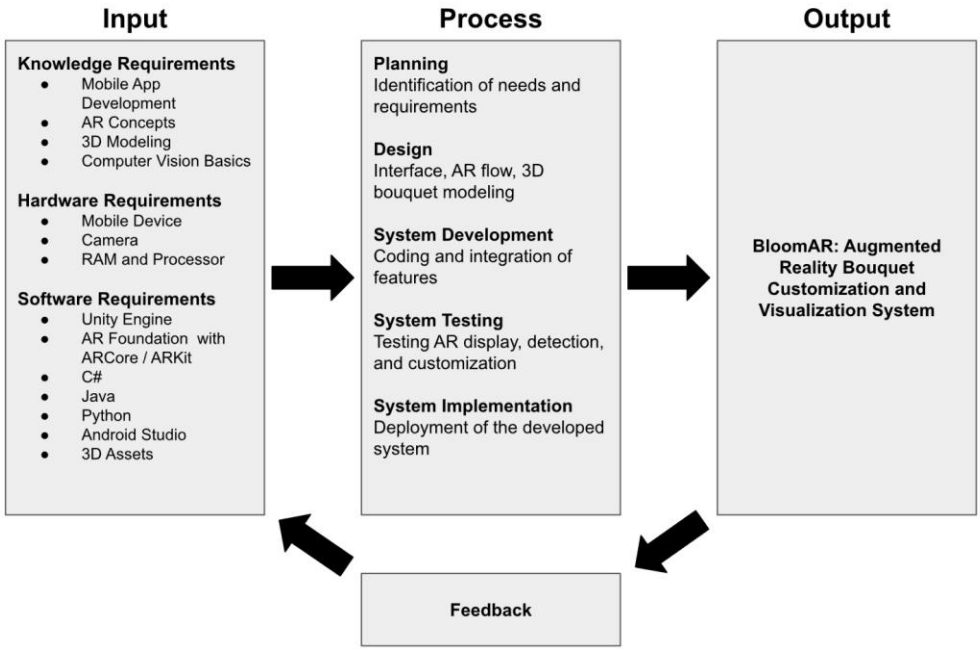


Figure 1: Conceptual Framework

The conceptual framework of BloomAR follows an input–process–output model with feedback integration.

The inputs include developer knowledge in mobile application development, augmented reality principles, 3D modeling, and computer vision fundamentals. Hardware inputs consist of an AR-supported mobile device with a camera and motion sensors.

Software inputs include Unity Engine, AR Foundation (with ARCore/ARKit support), C#, Android Studio, and 3D asset libraries.

Then the process stage follows systematic development phases, including planning, interface design, system development, algorithm implementation, testing, and deployment. During development, markerless AR tracking and feature-based detection algorithms are integrated to support real-time bouquet placement and interaction.

The output of the framework is the Bloom AR mobile application, which enables bouquet customization and real-time AR visualization. Feedback from users and evaluators is used to refine system features and improve performance, creating a continuous improvement loop.

This structured framework demonstrates how technical resources and user requirements are transformed into a functional AR bouquet visualization system.

Definitions of Terms

The following terms are defined operationally and technically to provide a clearer understanding of the concepts used in this study.

Application Logic. The programmed rules and functions that control how BloomAR processes user inputs, manages bouquet customization, and coordinates augmented reality visualization.

Augmented Reality (AR). A technology that overlays digital three-dimensional objects onto the real-world environment in real time using a mobile device camera.

AR Foundation. A Unity framework that integrates ARCore and ARKit to enable cross-platform augmented reality development.

AR Rendering. The real-time process of generating and displaying three-dimensional bouquet models within the physical environment through the device camera.

ARCore. Google's augmented reality platform for Android devices that provides motion tracking, environmental understanding, and light estimation.

ARKit. Apple's augmented reality framework that enables motion tracking, plane detection, and rendering on iOS devices.

Bouquet Customization. The process of selecting, modifying, duplicating, arranging, and adjusting digital floral elements such as flowers, colors, and styles within the BloomAR application.

C#. An object-oriented programming language used to implement application logic and AR interaction within the Unity environment.

Feature-Based Detection. A computer vision technique that identifies distinct visual points such as edges, corners, and textures from the camera feed to support surface detection and object placement in AR.

Markerless AR Tracking. A tracking method that allows virtual objects to be placed in a real-world environment without the use of printed markers, relying instead on environmental feature detection.

Mobile Application. A software program designed to run on smartphones or tablets, enabling users to access Bloom AR's bouquet customization and visualization features.

Rendering Algorithm. A computational process responsible for transforming digital 3D models into visual images displayed on the screen, including positioning, scaling, lighting, and shading.

Real Flower Detection. A system feature that uses the mobile device camera to analyze the visual characteristics of a physical flower and allow it to be referenced or duplicated within the application.

System Architecture. The structural design of BloomAR that defines how the user interface, camera processing, application logic, and AR rendering components interact.

System Implementation. The deployment and actual use of the completed BloomAR application on supported mobile devices.

System Testing. The evaluation process is used to verify that the application performs accurately, including AR tracking stability, customization functions, and rendering performance.

Three-Dimensional (3D) Assets. Digital models of flowers, ribbons, and bouquet elements used within the augmented reality environment.

User Interface (UI). The visual layout, controls, and interactive elements that allow users to navigate and operate the BloomAR application.

Unity Engine. A cross-platform development platform used to build the BloomAR application, manage 3D assets, and integrate augmented reality functionalities.

Feedback. Information gathered from users or testers to evaluate system performance and guide future improvements.

CHAPTER II

This chapter presents systems and studies related to the development of Bloom AR: An Augmented Reality Bouquet Customization and Visualization System. These references provided the researchers with foundational knowledge and technical insights necessary to conceptualize and implement the project. The section also outlines the technologies, tools, and methodologies that informed the system's design and functionality.

Related Systems

The study *"AI and AR Integration in E-Commerce Personalization"* by Putra and Hidayat (2018) explored the innovative combination of artificial intelligence and augmented reality in the field of online retail. Their system enabled users to virtually try cosmetic products while simultaneously receiving AI-driven recommendations tailored to their preferences and purchasing history. The methodology involved deploying AR interfaces on mobile devices and integrating machine learning algorithms to analyze user behavior and provide personalized suggestions. Results revealed that AR try-on features significantly increased user confidence, reduced hesitation in decision-making, and boosted purchase intention by allowing customers to visualize products before committing to a purchase. This research is highly relevant to BloomAR, as it demonstrates how AR can reduce uncertainty in decision-making and improve user satisfaction. BloomAR applies similar principles by enabling clients to preview bouquet designs in real time, thereby fostering confidence, minimizing revisions, and enhancing communication between florists and clients. Just as AR cosmetics try-on improved consumer trust in

product outcomes, BloomAR leverages AR visualization to ensure that floral arrangements meet client expectations prior to finalization.

And,*"Workflow Analysis for AR Customization Systems"* by Kim (2016) clarified the processes involved in developing interactive AR applications. The study utilized expert interviews and iterative prototyping to identify the critical stages of AR customization workflows, including input capture, parameterization, rendering, and feedback. Kim emphasized the importance of structuring these processes to ensure usability, efficiency, and scalability in AR systems. The research proposed a modular workflow model that could be adapted across different industries, highlighting how iterative refinement and user feedback loops are essential for system improvement. This methodological research is particularly useful for BloomAR, as it offers a framework for organizing its bouquet design pipeline. By adapting these workflow models, BloomAR ensures that users can smoothly transition from selecting event categories and floral elements to capturing real flowers, customizing duplicates, and previewing arrangements in AR mode. The emphasis on workflow clarity and iterative refinement directly informs BloomAR's development strategy, ensuring that the system remains user-friendly, responsive, and adaptable to evolving client needs.

Related Literatures

The study, *"Virtual and Augmented Reality: Trends, Application, Adoption, and Development in the Philippines"* by Panganiban, Menorca, Vinluan, and Rosales (2025), provided a comprehensive overview of VR and AR technologies in the country. This conference paper discussed the types, working principles, and applications of immersive

technologies across industries, including education, retail, healthcare, and design. The authors highlighted adoption barriers such as device limitations, inconsistent internet connectivity, and the need for user training, which often hinder widespread implementation. Despite these challenges, the study emphasized the growing interest in AR applications, noting that Filipino users are increasingly open to interactive and immersive experiences. The paper also underscored the importance of developing lightweight, mobile-friendly AR systems to ensure accessibility for a broader population. For BloomAR, these insights highlight the importance of optimizing the system for mobile platforms, ensuring compatibility with mid-range devices, and incorporating intuitive tutorials to support user onboarding. By addressing these adoption challenges, Bloom AR positions itself as an accessible and user-friendly application that can thrive in the Philippine market. Furthermore, the study's emphasis on AR's potential to transform traditional industries reinforces Bloom AR's vision of modernizing floral arrangement planning through immersive technology.

The article *"The Rise of an Augmented Nation: A Review of Augmented Reality Research in the Philippines"* by Motea (2024) synthesized the current state of AR research in the country, offering a broad perspective on its applications and future directions. The review highlighted the Philippines' dynamic digital landscape and the increasing adoption of AR among its tech-savvy population. It noted that AR is transitioning from a niche technology used primarily in gaming and entertainment to a transformative tool across education, commerce, healthcare, and design. The study emphasized that AR's growth is driven by both technological advancements and the demand for more interactive, user-centered experiences. This research is particularly relevant to BloomAR as it situates

the project within the broader national trend of AR adoption, emphasizing the potential for AR systems to modernize traditional industries such as floral arrangement and event planning. By aligning with this trajectory, BloomAR demonstrates how AR can be applied beyond conventional domains, offering a practical solution to a long-standing challenge in the floral industry helping clients visualize and customize bouquets with precision and creativity.

The systematic review *"Foreign Language Learning Using Augmented Reality Environments"* by Schorr, Plecher, Eichhorn, and Klinker (2024) examined the role of AR in language education by synthesizing findings from multiple empirical studies. The authors analyzed how AR technologies were integrated into classroom settings, mobile applications, and interactive learning platforms, identifying both benefits and challenges. Their review revealed that AR enhanced student engagement, improved retention of vocabulary and grammar, and provided immersive experiences that supported skill acquisition through contextualized learning. The study also highlighted challenges such as technical limitations, device compatibility, and the need for teacher training to maximize AR's effectiveness. Although the focus was on language learning, the research demonstrates AR's broader potential to improve user interaction and comprehension in diverse contexts. For Bloom AR, this study underscores the importance of designing an application that not only provides immersive visualization but also ensures accessibility and ease of use. Just as AR helped learners grasp abstract linguistic concepts more clearly, Bloom AR leverages AR to help clients conceptualize floral arrangements with precision and confidence, reducing ambiguity and enhancing satisfaction with final designs.

Ultimately, the article *"Enhancing Multi-Modal Perception and Interaction: An Augmented Reality Visualization System"* published in *Systems* (MDPI, 2023) proposed an AR visualization framework designed to improve human decision-making through enhanced perception and interaction. The study introduced a system that integrated multiple sensory inputs—such as visual, auditory, and tactile feedback—to create richer, more accurate visualizations. The authors demonstrated how multi-modal AR systems could support complex decision-making tasks by providing users with realistic and interactive representations of objects and environments. Their findings emphasized that AR's strength lies not only in visualization but also in its ability to engage multiple senses simultaneously, thereby increasing confidence and reducing errors in judgment. This research is directly applicable to BloomAR, as it underscores the importance of realistic rendering and interactive feedback in supporting confident decision-making during bouquet customization. By incorporating high-quality visual models and responsive interaction features, BloomAR ensures that clients can evaluate floral arrangements from multiple perspectives, make informed adjustments, and finalize designs with greater certainty.

TECHNICAL BACKGROUND

This section presents the technical foundation of Bloom AR: An Augmented Reality Bouquet Customization and Visualization System, detailing the development tools, platforms, and frameworks employed in the creation of its bouquet customization and AR visualization features. The researchers conducted an extensive review of relevant literature and technologies in the fields of mobile application development, augmented reality

integration, and real-time object detection to ensure the system's feasibility, reliability, and effectiveness.

Furthermore, the team examined existing AR-based customization systems to identify best practices in user-centered design, visual feedback, and interactive control mechanisms. This exploration informed the system's design decisions, particularly in enabling users to rotate, resize, and reposition bouquet elements with precision and ease. review emphasized the growing interest in AR technologies across education, retail, and design industries, noting that Filipino users are increasingly open to interactive and immersive experiences. For Bloom AR, these findings underscore the importance of optimizing the system for mobile platforms, ensuring compatibility with mid-range devices, and incorporating intuitive tutorials to support user onboarding. By addressing these adoption challenges, Bloom AR positions itself as an accessible and user-friendly application that can thrive in the Philippine market. Furthermore, Garcia's review highlights the potential of AR to transform traditional industries, a vision that Bloom AR embodies by modernizing floral arrangement planning and making it more interactive, efficient, and engaging.

SYNTHESIS OF RELATED STUDIES

The reviewed studies demonstrate that augmented reality enhances visualization, engagement, and user confidence across retail, education, and interactive systems. Existing AR applications primarily focus on product try-on, immersive learning, and general object placement. While these systems validate the effectiveness of AR for visualization and decision support, limited research addresses its application in floral customization and bouquet planning.

Most AR customization systems allow object visualization but do not integrate real-world object capture, digital duplication, and interactive arrangement within a single mobile platform. BloomAR addresses this gap by combining markerless AR tracking, feature-based detection, and real-time rendering algorithms to support interactive bouquet customization and visualization. By tailoring AR technology specifically to floral arrangement planning, the system contributes to a specialized solution that extends AR applications into the creative design domain.

CHAPTER III

DESIGN AND METHODOLOGY

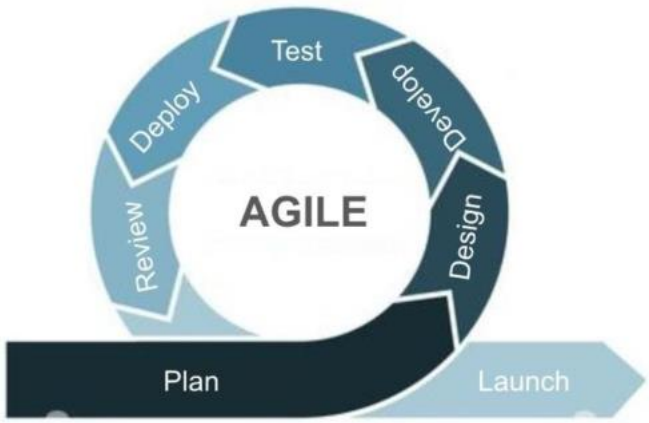
This study adopted a developmental research approach focused on the design, implementation, and evaluation of a mobile-based augmented reality system. The objective was to develop Bloom AR as a functional and user-centered bouquet customization and visualization application. The methodology involved requirement analysis, system design, algorithm integration, implementation, and user evaluation to ensure both technical reliability and usability.

Methods of the Study

The researchers adopted a developmental research approach, combining technical prototyping with iterative design evaluation. The study focused on designing, implementing, and refining a mobile-based AR system that enables users to select event-specific bouquet categories, customize floral elements, and preview arrangements through augmented reality. The methodology involved multiple phases: requirement gathering, interface design, system development, and usability testing. Data collection included user feedback, performance metrics, and observational analysis to assess the system's functionality, responsiveness, and user experience. The developmental nature of the study allowed for continuous improvement based on stakeholder input, ensuring that BloomAR met both technical standards and user expectations.

Software Development Life Cycle

This study used the Agile methodology because it supports flexible development and allows continuous improvements throughout the project. BloomAR requires frequent testing of its AR features, detection accuracy, and interface flow, which makes a traditional linear model less suitable. Agile made it possible to revise features as soon as user feedback was received, ensuring that each update responded to real needs and actual usage.



The development process was carried out through repeating sprint cycles that included planning, interface design, coding, and evaluation. These cycles encouraged regular refinement and quick adjustments whenever new requirements appeared. Input from florists, event planners, and general users was gathered during development to guide decisions and improve system usability.

The initial sprints focused on building the system's core features, such as bouquet category selection, real-world flower detection, virtual duplication of floral elements, and the AR-based preview. Unity served as the main development engine, while Vuforia

handled AR tracking and rendering. Android Studio supported mobile deployment, and the programming tasks were handled using C#, Java, and Python. These tools made it possible to create interactive controls, image recognition functions, and real-time rendering.

The Agile approach helped BloomAR evolve in a way that matched the conceptual framework. Inputs guided development, the process stages shaped improvements, and the output was strengthened through regular feedback. This method ensured that the system grew alongside the needs of users and the technical demands of AR-based bouquet customization.

Algorithms Used in the System

The Bloom AR system utilizes marker less augmented reality tracking supported by feature-based detection and real-time rendering algorithms to enable bouquet customization and visualization in a real-world environment.

Marker less AR Tracking

The primary algorithm used in the system is markerless AR tracking. This approach allows virtual bouquet models to be placed in the physical environment without requiring printed markers or QR codes. When the user activates the camera, the system scans the surroundings and detects flat surfaces where digital objects can be anchored.

This tracking method enables the bouquet to remain stable even when the user moves the device. The algorithm continuously updates the object's position relative to the environment, ensuring accurate alignment between the virtual bouquet and the real-world surface.

Feature-Based Detection

To support marker less tracking, the system relies on feature-based detection. This technique analyzes visual elements such as edges, corners, and texture differences captured by the camera. These detected features help the system understand spatial structure and determine where virtual objects can be safely positioned.

In Bloom AR, feature detection assists in identifying appropriate surfaces for bouquet placement and maintaining stable tracking during customization.

Rendering Algorithm

Rendering algorithms are responsible for generating and displaying the 3D bouquet models in real time. These algorithms process transformations such as scaling, rotation, duplication, and repositioning based on user interaction.

Whenever the user modifies the bouquet, the rendering engine updates the 3D model instantly. This ensures smooth visualization and realistic integration between digital flowers and the real environment.

System Development Tools

The development of Bloom AR made use of several software and hardware tools that supported the creation, testing, and deployment of its bouquet customization and augmented reality features. Unity 3D served as the main development environment where the interface, scene structures, and 3D assets were created and managed. It provided the scripting workspace used to handle user interactions and bouquet customization behaviors.

Vuforia SDK supported the AR capabilities of the system by enabling camera tracking, image detection, and the placement of virtual elements in real-world spaces. AR Foundation, AR Core, and ARKit worked alongside Unity to connect the mobile device's sensors and camera functions to the application's AR components. Android Studio was used during the deployment stage to prepare the application for Android devices and secure proper handling of mobile operations and system permissions. The programming of system functions was done using C#, Java, and Python. C# controlled the application logic inside Unity, Java supported mobile-side operations in Android Studio, and Python assisted during the early prototype testing stages when analyzing detection behavior.

Hardware tools were also essential in evaluating and running the AR system. An Android smartphone was used as a testing device since its camera, motion sensors, and processing capabilities allowed the AR features to function in real time. This device made it possible to assess detection stability, model rendering, and bouquet customization behavior during actual use. A development laptop or desktop computer with sufficient memory and processing power was required for building the system, editing assets, and running Unity's development environment. These hardware resources supported the entire development workflow and ensured that BloomAR could be built, tested, and optimized for mobile deployment.

System Design

This chapter presents the system design of Bloom AR, an augmented reality mobile application developed to support bouquet customization and visualization using real-world flower detection. The system design discusses the overall application flow, key system components, and the interaction between users and the augmented reality environment. It also covers the functional and non-functional requirements, as well as a detailed explanation of how the customization and AR visualization features operate within the system.

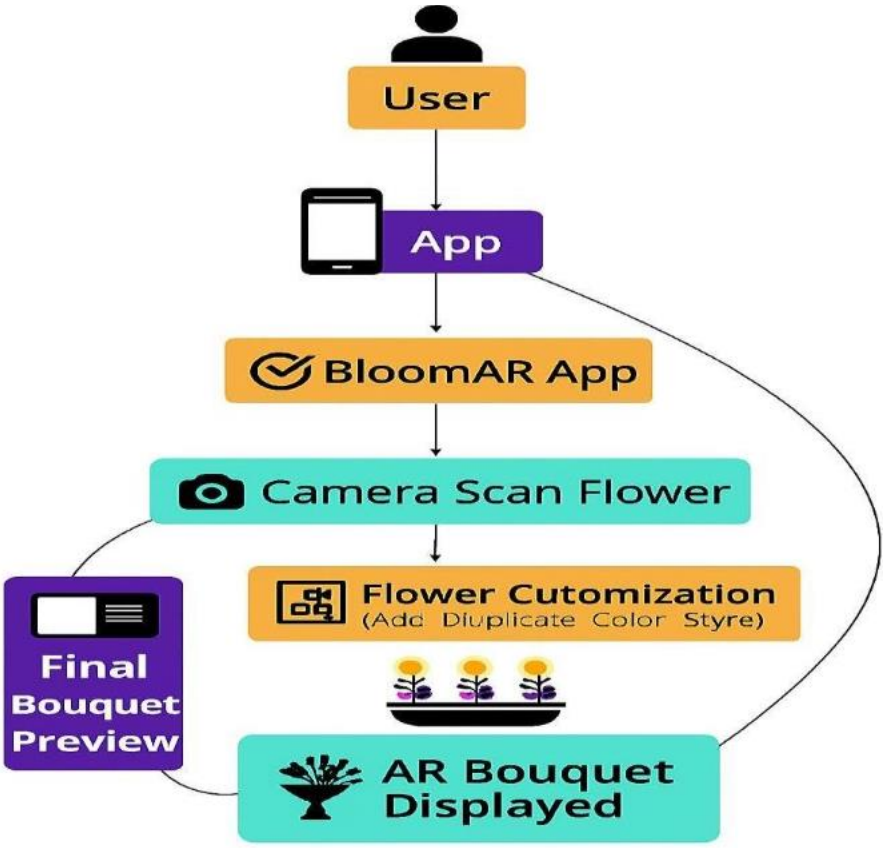


Figure 3. System Design

The system design diagram illustrates the overall workflow of BloomAR, showing how the user interacts with the application and how the system processes inputs to produce an augmented reality bouquet preview. The design highlights the step-by-step flow from user interaction to the final output, ensuring clarity in how the system functions internally.

The process begins with the user who initiates the interaction by accessing the application through a mobile device. Once the user opens the BloomAR app, the system prepares the necessary components, including the camera and augmented reality environment. This stage serves as the entry point where the application loads its features and allows the user to proceed with bouquet creation.

After launching the application, the system activates the camera scan flower function. At this stage, the user points the device's camera toward a real flower. The system uses its AR and detection to process visual data from the environment. This step allows the application to recognize the presence of a real flower, which becomes the base element for the digital bouquet. The scanned flower will be the reference for further customization within the system.

Once the flower is detected, the system moves to the flower customization stage. In this part of the process, the user is provided with various options to modify the bouquet. These options include adding more flowers, duplicating existing floral elements, changing colors, and adjusting styles. This stage allows users to experiment creatively and design a bouquet according to their preferences, even if only a limited number of physical flowers are available in the real world.

The system produces and shows the AR bouquet after completing the customization process. The customized bouquet is rendered in real time and overlaid onto the real-world view through the device's camera. The augmented reality display enables users to view the final arrangement as it would look in a real environment, which helps them visualize the bouquet design accurately.

The process concludes with the final bouquet preview, where the user can view the completed arrangement in augmented reality. This final output enables the user to evaluate the design, make final adjustments if needed, or use the preview as a reference for decision-making in floral arrangements and event planning. The diagram shows a looping flow that enables users to go back to previous stages, including customization for additional bouquet refinement through an interactive design approach.

Overall, the system design demonstrates how BloomAR integrates user input with camera-based object identification and customization features and AR displays into one functional system. The system design enables users to create flower bouquets in real time through its user-friendly interface, which allows flexible operational capabilities.

System Architecture

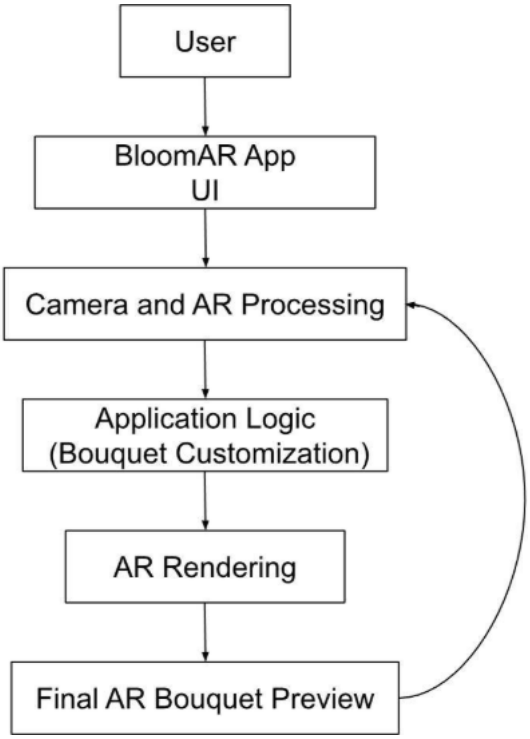


Figure 4: System Architecture

The figure illustrates the system architecture of the Bloom AR app. The process begins with the user, who interacts with the system through the Bloom AR app UI, to scan real flowers and customize bouquet elements. The camera and AR processing component capture, detect the flower, and prepare the real environment for AR display. The application logic manages customization features such as duplicating flowers, changing colors, resizing, and arranging the bouquet. These updates are sent to the AR Rendering component, which displays the customized 3D bouquet in real time. The output is shown as the Final AR Bouquet Preview, allowing users to view and refine their design continuously.

Context Flow Diagram

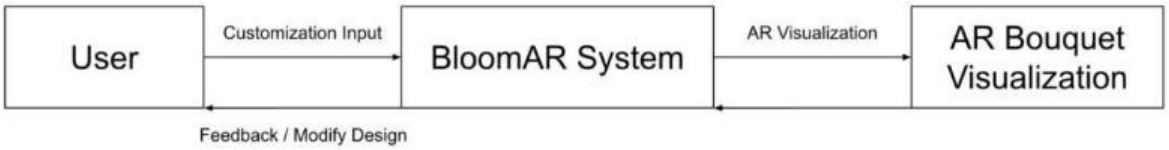


Figure 5: Context Flow Diagram

The diagram illustrates the interaction between the user, the BloomAR system, and the AR bouquet visualization. The user provides customization inputs such as flower selection, color variation, and arrangement preferences to the BloomAR System. The system processes these inputs and generates a real-time AR bouquet visualization that is displayed in the user's physical environment, and visual feedback from the AR visualization allows the user to modify and refine the bouquet design iteratively until the desired result is achieved.

Functional Requirements

Functional requirements constitute the specific capabilities, features, and operational functionalities of a mobile application. These detailed specifications delineate precisely what the application must achieve and how it should execute its functions to successfully meet its predefined goals and objectives. For user-centric applications, such as games, these requirements include a description of every user-facing feature to ensure clarity and proper implementation.

Table 1: Functional Requirements

Characteristics/Attribute	Description
User Access and Application Initialization	The system shall allow the User to launch and access the App to begin the bouquet creation process.
Flower Identification and Capture (Camera Scan Flower)	The system shall allow the user to access the mobile device camera to scan a real flower, enabling the app to identify and potentially use it as a base for the virtual bouquet.
Initial Bouquet Display (BloomAR App)	The system shall display the initial application interface and options, leading the user to either scan a flower or begin the customization process.
Customization (Flower Customization)	The system shall provide tools that allow the user to modify the virtual bouquet by: • Add: Including new types of flowers or foliage. • Duplicate: Copying existing flower elements within the arrangement. • Color: Changing the color of selected flower elements. • Style: Adjusting the arrangement style or density of the bouquet.
Augmented Reality Visualization (AR Bouquet Displayed)	The system shall render the customized 3D bouquet model in real-time using AR technology, displaying it aligned with the user's physical environment.

Final Review and Confirmation (Final Bouquet Preview)	The system shall present a final, static view of the fully customized bouquet design before saving or initiating the order process.

The functional requirements for the Bloom AR application define a clear, interactive workflow for augmented reality bouquet creation, beginning with User Access and Application Initialization, which allows immediate entry into the app to start the design process. The system then guides the user through the Initial Bouquet Display (Bloom AR App) interface, presenting options to either proceed to customization or utilize the Flower Identification and Capture (Camera Scan Flower) function, which allows the user to scan a real flower and use it as a foundation for the virtual bouquet. The core design is achieved through customization (flower customization), which provides comprehensive tools for modifying the virtual bouquet by adding new elements, duplicating existing ones, changing their color, and adjusting the arrangement style or density. After customization, the design is visualized via Augmented Reality Visualization (AR Bouquet Display), where the system renders the customized 3D model in real-time, aligned perfectly with the user's physical environment. The final step is the Final Review and Confirmation (Final Bouquet Preview), where the system presents a static view of the finished bouquet design for the user to confirm before saving or initiating the order process, ensuring a seamless journey from launch to final design.

Non-Functional Requirements

The Non-Functional Requirements for BloomAR: An Augmented Bouquet Customization and Visualization App ensure a high-quality user experience and optimal system performance. These requirements focus on the application's usability, compatibility, efficiency, reliability, maintainability, and security.

Table 2: Non-Functional Requirements

Attribute/Property	**Measures**
Usability	The system shall provide a user-friendly and intuitive interface suitable for first-time users. Customization options, flower libraries, and placement controls shall be clearly labeled and accessible with a maximum of three taps from the main screen.
Compatibality	The system shall be compatible with mobile devices that support ARKit (iOS) and ARCore (Android) and maintain a stable frame rate of at least 30 frames per second (FPS) during AR rendering.
Efficiency	The system shall ensure that AR-mode launch and surface detection take no longer than 5 seconds on supported devices. The system shall optimize the use of device resources, minimizing battery consumption during continuous AR operation.
Reliability	The system shall operate consistently without unexpected crashes or loss of AR surface tracking during normal use. The application shall recover and re-establish the AR session if tracking is temporarily lost within 2 seconds.

Maintainability	The system shall be developed using modular architecture (e.g., MVVM or MVC) and well-documented code to allow new flower models, colors, and customization features to be added with minimal structural changes.
Security	The system shall only access the mobile device's camera and storage with the user's explicit permission. Any user-saved designs or contact information must be stored securely using industry-standard encryption protocols.

The non-functional requirements for the BloomAR application are crucial for ensuring a high-quality user experience and robust system performance, beginning with usability, which requires a user-friendly interface where all customization tools and controls are accessible within a maximum of three taps from the main screen. Compatibility mandates that the system work across both ARKit (iOS) and ARCore (Android) devices while maintaining a stable frame rate of at least 30 FPS. Efficiency (performance) focuses on speed, limiting the AR-mode launch and surface detection time to no more than 5 seconds, and optimizing resource use to minimize battery drain. Reliability ensures consistent operation without crashes and specifies that the system must recover from temporary AR surface tracking loss within 2 seconds. Maintainability guarantees the system's longevity and adaptability by requiring a modular architecture (e.g., MVVM or MVC) and well-documented code for easy feature additions. Finally, security protects the

user by mandating explicit permission for camera and storage access and requiring all user data and saved designs to be stored using industry-standard encryption protocols.

Table 3: Hardware Requirements in Developing the System

Equipment	Type/Specifications
Development Computer /Laptop	Intel Core i5 or higher, 8 GB RAM or higher, dedicated or integrated GPU
Android Smartphone (Test Device)	Android OS 9.0 or higher, Rear Camera, 4 GB RAMminimum
Camera (Built-in Mobile Camera)	At least 8 MP resolution
Storage Device	256 GB HDD/SSD or higher

In table 3, the researchers present the hardware requirements needed for the development of the BloomAR system. These hardware components are necessary to support mobile application development, augmented reality processing, and three-dimensional rendering. A development computer or laptop with sufficient processing power and memory is required to run software tools such as Unity and Android Studio smoothly. An Android smartphone is also needed as a testing device to evaluate the system's AR features, real-flower detection, and real-time bouquet visualization. The built-in camera is essential for capturing real flowers and providing the live view used in augmented reality, while adequate storage ensures that project files, 3D assets, and application builds can be properly managed during development.

Table 4: Software Requirements for Developing the System

Software Requirements	Specifications
Unity Engine	Unity 2021 or later
Vuforia SDK	Latest Stable Version
AR Foundation	Latest Stable Version
Android Studio	Latest Version
Programming Languages	C#, Java, Python
3D Asset Libraries	Free or Licensed Assets

This table shows the software requirements used in developing the BloomAR system. The listed software tools support the creation, integration, and deployment of the application's augmented reality and customization features. Unity Engine serves as the main development platform for building the user interface and managing 3D bouquet models, while the Vuforia SDK and AR Foundation enable flower detection, camera tracking, and AR rendering. Android Studio is used for compiling, testing, and deploying applications on Android devices. Programming languages such as C#, Java, and Python are utilized to handle system logic, mobile operations, and prototype testing, ensuring that the system functions correctly and efficiently.

Table 5: Software Specification in Using the System

Software	Specification
Android Operating System	Android 9.0 or higher
Bloom AR Mobile App	Installed APK
ARCore Support	Enabled
Internet Connection	Optional

The software specifications for using Bloom AR define the minimum system requirements needed for end users to operate the application effectively. The system is designed to run on Android devices with version 9.0 or higher to ensure compatibility with modern AR libraries and device sensors. The Bloom AR mobile application must be properly installed on the device to access bouquet customization features and augmented reality visualization.

ARCore support is a critical requirement, as it enables accurate motion tracking, environmental understanding, and real-time rendering of digital bouquets. Without ARCore compatibility, the system would not be able to anchor virtual objects correctly in the physical environment. While Bloom AR can operate offline for most customization and visualization features, an internet connection may be required for application updates, asset downloads, or system improvements.

Table 6: Hardware Specification in Using the System

Hardware	Specification
Android Smartphone	Android 9.0+, 4 GB RAM
Camera	Rear Camera (8 MP or higher)
Processor	Octa-core or equivalent
Sensors	Gyroscope and Accelerometer
Storage	At least 500 MB free space

The hardware specifications for using BloomAR focus on ensuring a smooth and stable user experience during augmented reality interaction. An Android smartphone with at least 4 GB of RAM is required to handle real-time rendering of three-dimensional bouquet models and interactive controls. Adequate memory prevents lag and ensures smooth transitions when users rotate, resize, or reposition bouquet elements.

The rear camera plays a vital role in capturing real flowers and displaying the AR environment. A higher-resolution camera improves detection accuracy and enhances the realism of the AR preview. The device processor must be capable of handling multiple tasks simultaneously, including camera input processing, AR tracking, and rendering of 3D models.

Motion sensors such as the gyroscope and accelerometer are essential for detecting device movement and maintaining proper alignment between virtual bouquets and the real-world environment. Sufficient storage space is also necessary to store application files,

temporary data, and saved bouquet designs. Together, these hardware components ensure that users can fully experience BloomAR's features without performance issues.

System's Testing

The researchers supplied an evaluation questionnaire once the testing strategies were completed to give the BloomAR: Augmented Reality Bouquet Customization and Visualization App a more thorough evaluation. The formulation of the questionnaire was based on ISO/IEC 29110. This ISO/IEC standard defines five (5) key characteristics that describe software quality, which will be adapted to assess the specific performance of the AR application:

- Functionality: It includes verifying seamless surface detection and accurate application of customization options.
- Reliability: This will assess the ability of the BloomAR app to function under stated conditions for a specified period.
- Usability: This evaluates how easily users can explore and understand the BloomAR app.
- Efficiency: This means the BloomAR app responds precisely and effectively to the commands.
- Maintainability: It indicates that the BloomAR app is reliable and simple to manage.

Table 7: Likert Scale

Rating	Range	Equivalent
4	3.50-4.00	Strongly Agree
3	2.50-3.50	Agree
2	1.50-2.50	Disagree
1	1.00-1.50	Strongly Disagree

For this study, statistical analyses, frequency, and percentage were used. The outcomes of the system evaluation are determined using weighted means and distributions for data presentation. Below are the evaluation criteria used by the researchers.

Table 8: System Evaluation

Features	4 Strongly Agree	3 Agree	2 Disagree	1 Strongly Disagree
A. Functionality				
The system accurately detects real flowers using the device's camera.				
The system allows users to customize bouquets by adding, duplicating, resizing, and changing flower colors.				
The system correctly displays the customized bouquet in augmented reality mode.				
The system performs its intended features without errors during use.				
B. System Usability				
The system interface is easy to understand and navigate.				
All buttons, icons, and controls perform their intended functions.				

The AR viewing controls allow easy rotation, resizing, and repositioning of the bouquet.				
The system can be used with minimal guidance or instructions.				
C. Efficiency				
The system allows users to design bouquet arrangements faster.				
The system reduces the need for multiple physical flower samples.				
The system provides an efficient way to visualize bouquet designs before actual production.				
D. Reliability				
The system performs consistently during repeated use.				
The AR visualization remains stable during proper lighting conditions.				
The system performs its functions correctly without crashing.				
E. Maintainability				
The system structure allows easy modification of bouquet elements and features.				
The system can be updated or improved without affecting its core functions.				
The system design supports future enhancements and feature additions.				

Project Team and Their Responsibilities

The proponents of the study consist of two (2) members, each assigned specific roles that contribute to the development of BloomAR: An Augmented Reality Bouquet Customization and Visualization App, ensuring the project's success. Their responsibilities are outlined below:

Table 9: Project Team and their Responsibilities

MEMBERS NAME	RESPONSIBILITY
	Document Writer Application Designer Programmer
	Programmer Document Writer Data Researcher

Programmer

The programmer plays the crucial role of writing and managing the code necessary to implement the app's core features. In BloomAR, the programmer ensures that the AR surface detection and tracking algorithms function correctly, allowing the virtual bouquet to be stably anchored and scaled within the user's physical environment. They also handle the integration of the 3D rendering engine and customization logic.

Application Designer

The application designer is responsible for structuring the overall system design of BloomAR, ensuring the architecture is robust, modular, and supports future scalability. This includes designing the flow between the customization interface and the AR

visualization mode, organizing the app's layout, and ensuring all required AR and customization features are logically integrated for easy navigation.

Data Researcher

The data researcher gathers necessary information about floral design, including authentic flower species specifications, realistic color palettes, traditional bouquet arrangements, and foliage types. This research is crucial for ensuring that the app accurately represents authentic floral options and provides users with realistic and high-fidelity 3D assets for customization.

Document Writer

The document writer is responsible for recording all updates, modifications, and corrections related to the development of BloomAR. This role is essential for maintaining organized technical and procedural documentation, which includes AR algorithm adjustments, customization logic changes, system improvements, and ensuring smooth maintainability.

Schedule and Timeline

Table 8: Schedule and Timeline

The researchers' schedule and timeline, as presented in the Gantt chart below, show the weekly activities undertaken in the development of BloomAR: An Augmented Reality Bouquet Customization and Visualization System. The Gantt chart outlines the sequence and duration of tasks such as thesis proposal preparation, system planning, design, programming, testing, documentation, and implementation. By using the Gantt chart, the researchers were able to organize project activities, monitor progress, and ensure that each

phase of development was completed according to schedule. The chart also serves as a guide for tracking ongoing tasks, particularly those still in progress during the second week of December, and for adjusting timelines as needed to support the successful completion of the system.

*Legend: 1.00-1.50 = Strongly Disagree; 1.50-2.50 = Disagree; 2.50-3.50 = Agree; 3.50-4.00 = Strongly Agree*Table 15 illustrates the assessment findings in terms of the application's efficiency. The evaluation obtained a composite mean of 3.91, with a verbal interpretation of "Strongly Agree," indicating that the developed mobile game application generates accurate results. and discuss the efficiency of the developed mobile game application. Also provided in this section, the summary of recommendations for the enhancement of the game, made by the researchers based on the data gathered during the development of the mobile game application.developed game exhibited smooth gameplay with minimal delays and responsive controls. Survey results showed that the majority of respondents strongly agreed that the game operated efficiently.he difficulty curve was well-balanced, ensuring an engaging yet manageable challenge. Various playstyles were accommodated, and the progression system effectively maintained player interest.Kusina Craze successfully achieved its goal of providing an engaging and educational gaming experience centered on Filipino cuisine.developed game is well-designed, reliable, and fun to play, as indicated by the positive feedback from respondents.for the future researchers to incorporate more Filipino dishes and introducing additional interactive cooking methods, such as grilling, steaming, and frying, to diversify the gameplay experience..